Frozen DINOv3 Features for Martian Terrain Segmentation: Layer-wise Linear Probing on AI4Mars

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1. INTRODUCTION

Autonomous rovers need to recognize the terrain they drive on (soil, bedrock, sand, big rock), but expert labels of Mars are scarce and costly. DINOv3 [1] and similar self-supervised models are pretrained on web images, so it is unclear how well their frozen features transfer to a very different domain: grayscale imagery, repetitive textures, unusual illumination. We study this transfer on AI4Mars [2] with a minimal linear head and then with heavier heads, and ask (i) how well frozen DINOv3 features segment Martian terrain, (ii) how much of the result depends on pretraining (against a frozen random-init network, which is a lower bound), (iii) which layers carry the useful information and (iv) how much a larger head or fine-tuning adds. Published AI4Mars baselines [3,4] train convolutional or transformer segmenters end to end; none of the seven AI4Mars baseline papers we reviewed uses a frozen foundation backbone (MANTLE [7] does, for binary boulder segmentation on its own MSL data). The closest work, Goh et al. [6], pretrains a backbone without labels (contrastive loss) and fine-tunes it with few labels (91.1 % pixel accuracy with 161 images); we keep an off-the-shelf web-pretrained backbone frozen and report mIoU.

2. METHODOLOGY

Data. We use the Curiosity NavCam subset of AI4Mars [2] at 512×512 px; unlabeled pixels are ignored. Training uses 15,564 crowd-labelled images; layers and epochs are chosen on a validation split of 500 training-set images held out by acquisition time; the official 322-image expert test set is used only to confirm, in its three variants M1, M2 and M3 (agreement of at least 1, 2 or 3 experts), reported separately and scored at native label resolution.

Model. DINOv3 ViT-B/16 [1] (86 M parameters) is kept frozen. Patch features are read at blocks 2, 4, 6, 8, 10, 12 and after the final normalization (Fig. 1), and the same linear probe (batch normalization and a 1×1 convolution, 4,612 trainable parameters) is trained on each one [5]. We also train a decoder fusing blocks 3, 6, 9 and 12 (3.5 M parameters) and fine-tune the last 2 or 4 blocks (14 M / 28 M), all with three seeds.

Evaluation. We report mIoU over the four classes with 95 % paired bootstrap confidence intervals over images; a difference counts as real if its interval excludes 0 and $|\Delta| \geq 0.01$. Six hypotheses and their decision rules were written down before the final run; unsettled hypotheses are reported as inconclusive.

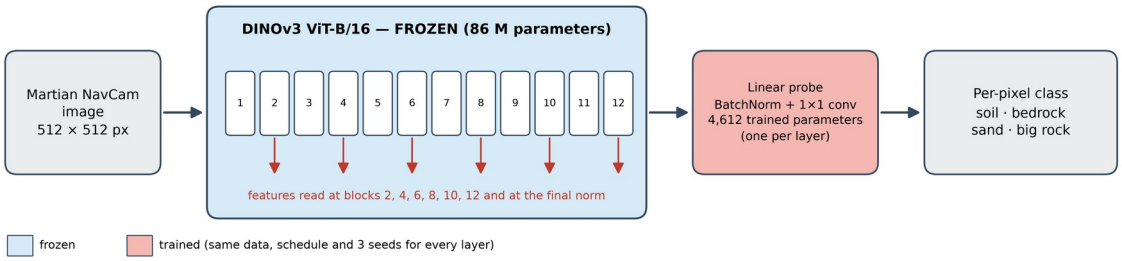


Fig. 1. Frozen DINOv3 ViT-B/16 with one trained linear probe per tapped layer.

3. RESULTS AND DISCUSSION

Transfer. A linear probe on the final layer reaches mIoU 0.677 / 0.725 on test M1 / M2 (0.651 on validation); always predicting the majority class gives 0.115 / 0.123 on M1 / M2 (0.125 on validation). M3: 0.827 (see Failures).

Heavier heads add little. The decoder reaches 0.684 / 0.735 and fine-tuning the last 4 blocks 0.687 / 0.737: at most +0.012 over the probe (decoder minus probe on M1: +0.007, 95 % CI [−0.001, +0.019]). 250 labelled images give 99 % of the probe's M1 mIoU obtained with 4,000.

Pretraining matters. A frozen random-init ViT-B/16 reaches 0.264 on M1 (DINOv3: +0.413). DINOv2 B/14 reaches 0.664; DINOv3 is +0.013 higher on M1 (95 % CI [+0.002, +0.025]) but not clearly on M2;

matching the token grid (DINOv2 at 448 px) leaves the gap unchanged (+0.014 on M1, 95 % CI [+0.003, +0.026]). All backbones here use N = 2,000 images, so each gap is between equal-N probes; DINOv3 at N = 2,000 (0.677) matches the full-data probe, so +0.413 holds for it too. A frozen random ViT is only a lower bound, not a from-scratch end-to-end control (not run).

Published baselines. The probe matches the 512-px published models and 0.023 / 0.015 below the best (UPerNet Swin-S at 1,024 px [3]) on M1 / M2; fine-tuning the last 4 blocks is 0.013 / 0.003 below, with 4.6 k–28 M trainable parameters instead of 58.8–81.2 M (Table 1). Indicative only: published values are single runs on another split.

Layers. mIoU rises with depth; block 6 gives 98 % of the final M1 value. No intermediate layer beats the last one: best on validation (block 10): −0.010 on M1 (95 % CI [−0.021, +0.001]).

Label borders. Excluding an 8 px band around label edges raises every head by ≈ 0.025 mIoU on M1 (probe 0.677 → 0.702), consistent with (not proof of) label noise at borders.

Failures. Big rock is the hardest class (validation IoU 0.24 against 0.62–0.88 for the others; 0.13 / 0.12 on test M1 / M2; 69 % of its pixels go to bedrock). It appears in only 5 of the 322 images of M3 (77,698 px): every model finds 96–99 % of them and they differ in false alarms (probe 109 k px, decoder 16 k px), which drives the M3 gap (0.83 vs 0.93); without big rock all score 0.96–0.97, so M3 cannot rank models on this class.

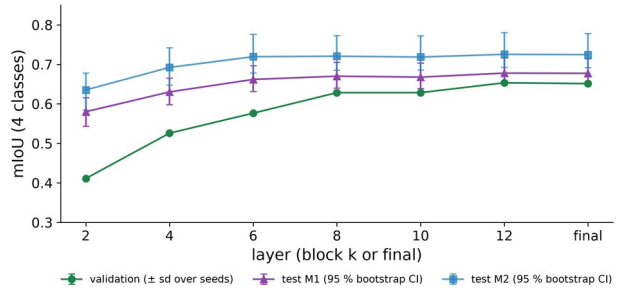


Fig. 2. Probe mIoU per layer of frozen DINOv3 (validation ± sd, 3 seeds; test with 95 % bootstrap CI).

Table 1. Test mIoU, mean ± sd over 3 seeds, versus published models. M3 is dominated by big-rock false alarms.

Method	M1	M2	M3	Trainable
Majority class	0.115	0.123	0.128	–
Probe, block 6	0.662 ± 0.001	0.719 ± 0.001	0.872 ± 0.007	4.6 k
Probe, final norm	**0.677 ± 0.001**	**0.725 ± 0.002**	**0.827 ± 0.018**	**4.6 k**
Decoder (blocks 3, 6, 9, 12)	0.684 ± 0.001	0.735 ± 0.001	0.931 ± 0.006	3.5 M
Last 4 blocks fine-tuned	0.687 ± 0.002	0.737 ± 0.002	0.927 ± 0.021	28 M
DINOv2 B/14, probe (N = 2,000)	0.664 ± 0.003	0.714 ± 0.004	0.798 ± 0.010	4.6 k
Random-init ViT, probe (N = 2,000)	0.264 ± 0.011	0.272 ± 0.012	0.272 ± 0.015	4.6 k
DeepLabV3+ R101, 512 px [3]	0.67	0.72	0.82	58.8 M
UPerNet Swin-S, 1,024 px [3]	0.70	0.74	0.94	81.2 M

4. CONCLUSIONS

Frozen DINOv3 features transfer to Martian terrain without adapting the backbone: a linear probe is within 0.025 mIoU of the best published end-to-end model on M1 and M2, heavier heads add at most 0.012, pretrained features matter (a frozen random ViT is far lower, as a lower bound), and no intermediate layer beats the last one.

Limitations. One backbone, one rover, three seeds and 322 test images per variant; only one of six pre-specified hypotheses was supported (H4: a 3.5 M decoder does not beat the probe) and the rest are inconclusive; M3 cannot rank models on big rock; the baseline comparison is indicative; no from-scratch control.

Next steps. Rerun a published baseline under our protocol, add a from-scratch control, other rovers. Code: https://github.com/gonzaquis29/DinoV3-AI4Mars.

Message for discussion. How much of the M1/M2 ceiling is label noise? How to evaluate rare classes?

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